ECE 3317
Applied Electromagnetic Waves

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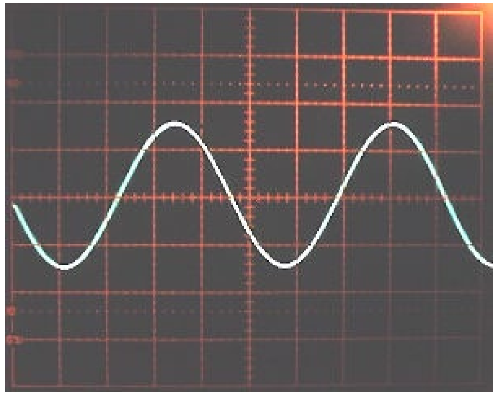
Fall 2025



Notes 2
Complex Vectors

Adapted from notes by Prof. Stuart A. Long

Notation

Circuit quantities:

- $v(t)$ is a time-varying function.
- V is a phasor (complex number).

Note:
"Handscript SF" font is used for time-domain vector quantities.
(This font has been placed on Canvas for you.)

Field quantities:

- $\underline{\mathcal{E}}(t)$ is a time-varying vector function.
- $\underline{E}$ is a phasor vector (complex vector).
- $\mathcal{E}_x(t)$ is a time-varying component of a vector function.
- E_x is a phasor component of a vector function.

Appendix B in the Hayt & Buck book discusses units in some detail.

Complex Numbers

$$c = a + j\,b = |c|\,e^{j\phi}$$

Real part | Imaginary part | Magnitude | Phase (always in <u>radians</u>)

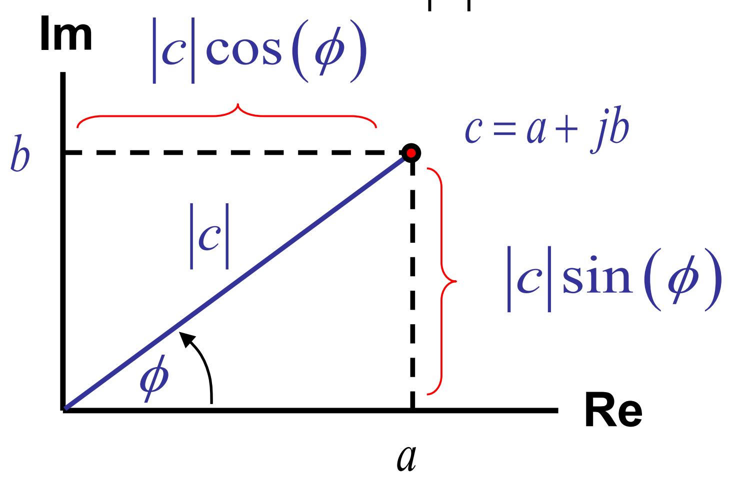
$$j = \sqrt{-1}$$

Note: a, b, ϕ are real.

Euler's identity allows us to write the polar form.

Polar form:

$$|c| = \sqrt{a^2 + b^2}$$



Euler's identity:

$$e^{j\phi} = \cos\phi + j\sin\phi$$

$$c = |c|\left(\cos\phi + j\sin\phi\right)$$

$$a = |c|\cos\phi$$

$$b = |c|\sin\phi$$

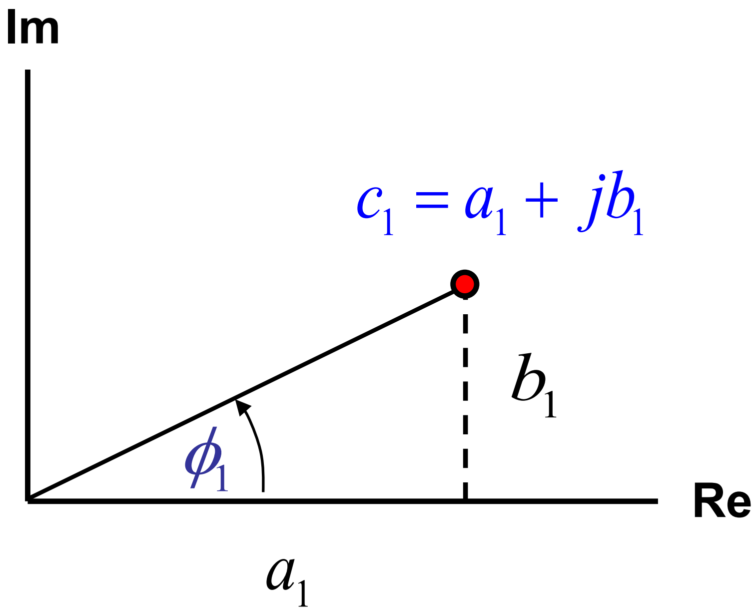
Note: The angle ϕ is measured positive going counterclockwise.

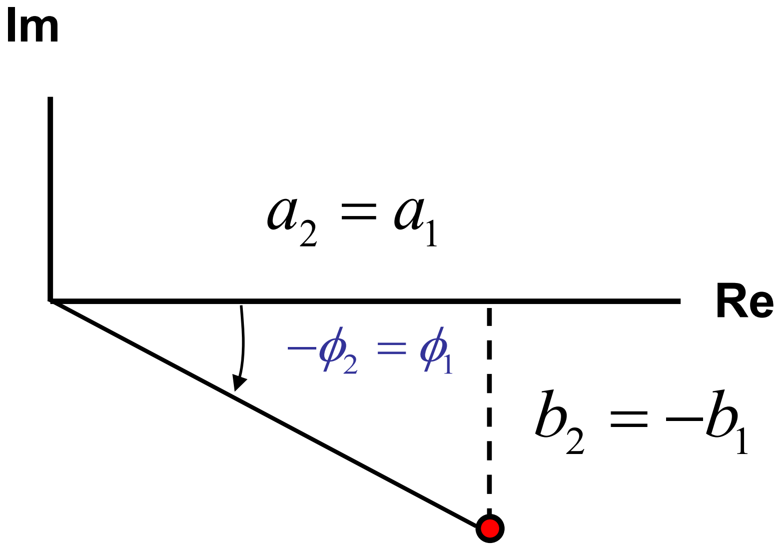
Complex Numbers (cont.)

Complex conjugate

$$c_1 = (a_1 + jb_1) = |c_1| e^{j\phi_1}$$

$$c_2 = c_1^* = (a_1 + jb_1)^* \equiv a_1 - jb_1 \; = |c_1| e^{j(-\phi_1)} \quad (\phi_2 = -\phi_1)$$





$$c_2 = a_2 + jb_2 = c_1^* = a_1 - jb_1$$

Complex Algebra

$$c_1 = a_1 + jb_1 = |c_1| e^{j\phi_1}$$
$$c_2 = a_2 + jb_2 = |c_2| e^{j\phi_2}$$

Addition

$$c_1 + c_2 = (a_1 + a_2) + j(b_1 + b_2)$$

Subtraction

$$c_1 - c_2 = (a_1 - a_2) + j(b_1 - b_2)$$

Complex Algebra (cont.)

$$c_1 = a_1 + jb_1 = |c_1| e^{j\phi_1}$$
$$c_2 = a_2 + jb_2 = |c_2| e^{j\phi_2}$$

Multiplication

$$(c_1)(c_2) = (a_1 + jb_1)(a_2 + jb_2) = (a_1a_2 - b_1b_2) + j(a_1b_2 + a_2b_1)$$
$$(c_1)(c_2) = |c_1||c_2| e^{j(\phi_1 + \phi_2)}$$

Division

$$\frac{c_1}{c_2} = \frac{a_1 + jb_1}{a_2 + jb_2} = \frac{a_1 + jb_1}{a_2 + jb_2}\left(\frac{a_2 - jb_2}{a_2 - jb_2}\right) = \frac{(a_1 + jb_1)(a_2 - jb_2)}{a_2^2 + b_2^2} = \frac{(a_1a_2 + jb_1b_2) + j(b_1a_2 - a_1b_2)}{a_2^2 + b_2^2}$$
$$\frac{c_1}{c_2} = \frac{|c_1|}{|c_2|} e^{j(\phi_1 - \phi_2)}$$

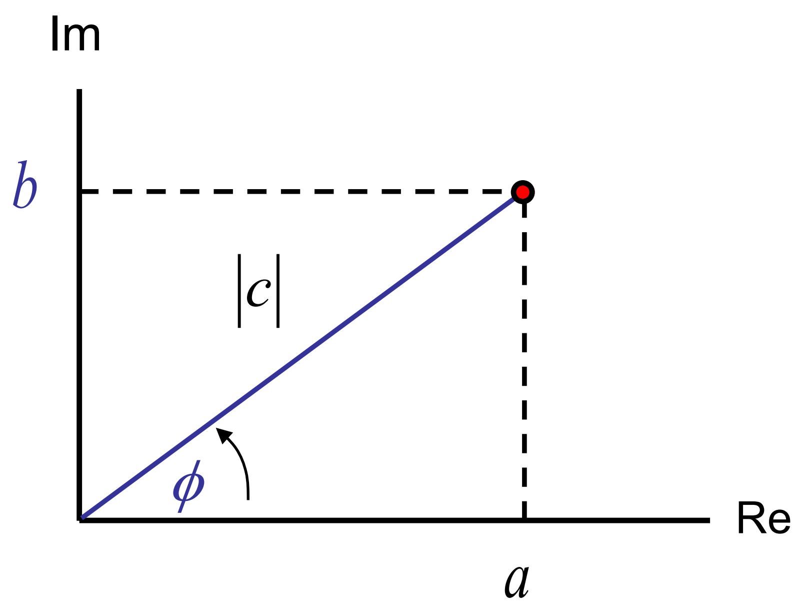
Square Root

Principal square root of a complex number (denoted by radical sign):

$$c = |c|\, e^{j\phi}$$

$$\sqrt{c} = \left(|c|\, e^{j\phi}\right)^{1/2} \qquad -\pi < \phi \le \pi$$ (the principal branch)

$$\sqrt{|c|}\; e^{j(\phi/2)}$$



Note:
For a positive real number x, the principal square root is positive:

Example: $\sqrt{4} = 2$

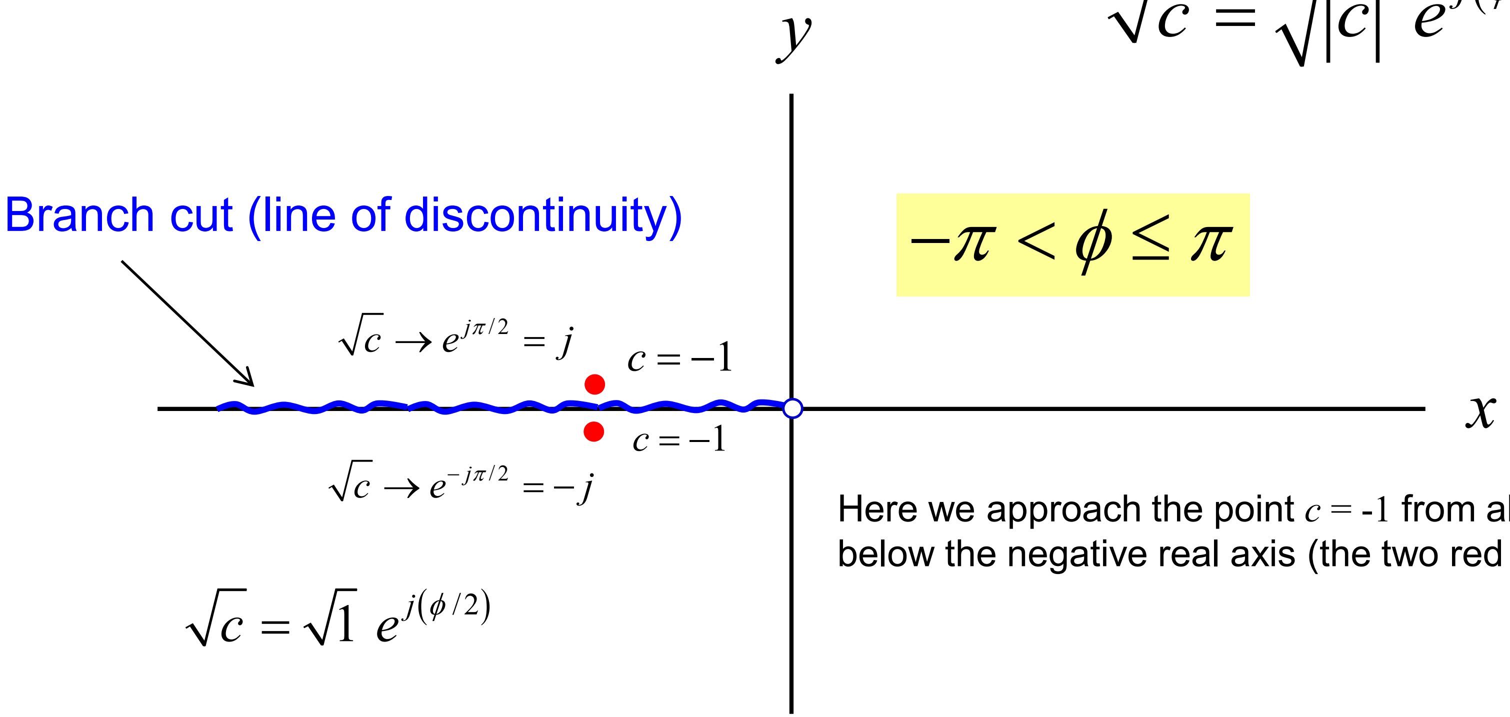
Square Root

$$c = |c| e^{j\phi}$$

Illustration of principal square root:

$$\sqrt{c} = \sqrt{|c|}\; e^{j(\phi/2)}$$

$$-\pi < \phi \le \pi$$



$$\sqrt{c} = \sqrt{1}\; e^{j(\phi/2)}$$

Here we approach the point $c = -1$ from above or below the negative real axis (the two red dots).

Note: Matlab uses the principal square root: $\sqrt{-1} = +j$

Square Root (cont.)

Property of principal square root:

$$-\pi < \phi \le \pi$$

$$\Downarrow$$

$$-\pi/2 < \phi/2 \le \pi/2$$

$$\Downarrow$$

$$\boxed{\mathrm{Re}\sqrt{c} \ge 0}$$

$$c = |c|\,e^{j\phi}$$

$$\sqrt{c} = \sqrt{|c|}\;e^{j(\phi/2)}$$

$$-\pi < \phi \le \pi$$

Examples :

$$\sqrt{4} = 2$$

$$\sqrt{j} = \sqrt{1e^{j\pi/2}} = \sqrt{1}\,e^{j\pi/4} = \frac{1+j}{\sqrt{2}}$$

Square Root (cont.)

$$c = |c| e^{j\phi}$$

General square root of a complex number (two values):

Denotes the angle of the principal branch

$$\begin{aligned} c^{1/2} &= \left(|c| e^{j\phi}\right)^{1/2} \\ &= \left(|c| e^{j(\phi_p + 2\pi n)}\right)^{1/2} \quad \left(-\pi < \phi_p \le \pi, \;\; n \text{ is any integer}\right) \\ &= \sqrt{|c|}\; e^{j(\phi_p/2 + 2\pi n/2)} \\ &= \left(\sqrt{|c|}\; e^{j\phi_p/2}\right) e^{jn\pi} \\ &= \pm\sqrt{|c|}\; e^{j\phi_p/2} \quad \left(+ \text{ for } n \text{ even, - for } n \text{ odd}\right) \\ &= \pm\sqrt{c} \end{aligned}$$

Examples :

$$4^{1/2} = \pm 2$$

$$j^{1/2} = \pm\left(\frac{1+j}{\sqrt{2}}\right)$$

The general square root has two possible values.

Time-Harmonic Quantities

$$v(t) = A\cos(\omega t + \phi)$$

Amplitude

Angular Frequency

Phase

Note: A, ω, ϕ are real.

$$\boxed{f = \frac{\omega}{2\pi}}$$ Frequency [Hz]

$$\boxed{T = \frac{1}{f}}$$ Period [s]

From Euler's identity:

$$v(t) = \mathrm{Re}\left\{A\,e^{j(\phi+\omega t)}\right\} = \mathrm{Re}\left\{\left(A\,e^{j\phi}\right)e^{j\omega t}\right\}$$

Define the phasor : $V \equiv A\,e^{j\phi}$

We then have:

$$v(t) = \mathrm{Re}\left\{V e^{j\omega t}\right\}$$

Note:
The phase angle of the sinusoidal wave is the same as the phase angle of the complex number (phasor). The amplitude of the sinusoidal wave is the same as the magnitude of the complex number (phasor).

Time-Harmonic Quantities (cont.)

$$v(t) = A\cos(\omega t + \phi)$$

$$\begin{cases} V \equiv A e^{j\phi} & \text{going from time domain to phasor domain} \\ v(t) = \mathrm{Re}\left\{V e^{j\omega t}\right\} & \text{going from phasor domain to time domain} \end{cases}$$

Time-domain $\Leftrightarrow$ Phasor domain

$$v(t) \Leftrightarrow V$$

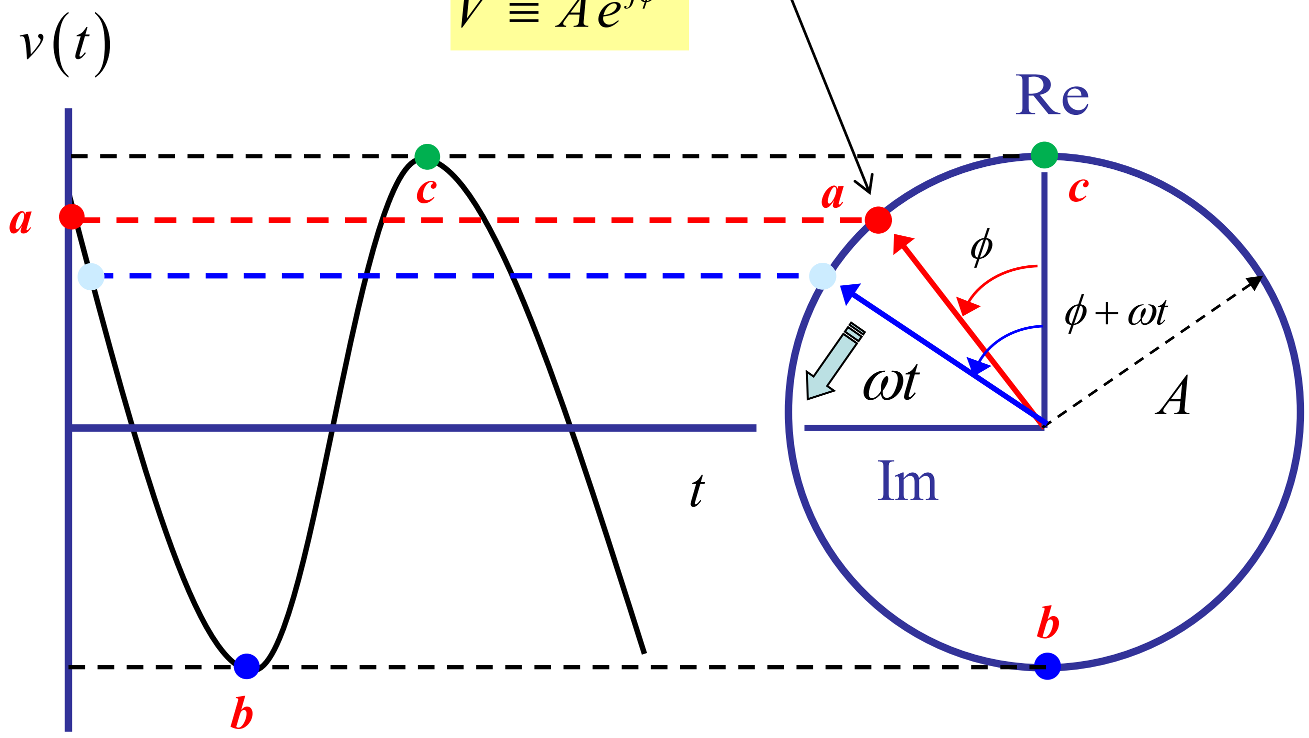
Time-Harmonic Quantities (cont.)

$$v(t) = A\cos(\omega t + \phi)$$

$$v(t) = \mathrm{Re}\left\{V e^{j\omega t}\right\}$$
$$= \mathrm{Re}\left\{A e^{j\phi}\, e^{j\omega t}\right\}$$

The complex number V

$$V \equiv A e^{j\phi}$$



Time-Harmonic Quantities (cont.)

$$v(t) \Leftrightarrow V$$

Note:

$$u(t) + v(t) \Leftrightarrow U + V$$

This assumes that the two sinusoidal signals are at the same frequency.

$$\frac{\partial}{\partial t} v(t) \Leftrightarrow j\omega V$$

There are no time derivatives in the phasor domain!

However:

$$u(t)\,v(t) \not\Leftrightarrow UV \quad \left(\text{i.e., } u(t)\,v(t) \neq \mathrm{Re}\left(UVe^{j\omega t}\right)\right)$$

All phasors are complex numbers, but not all complex numbers are phasors!

Complex Vectors

Vectors in the Phasor Domain

$$\underline{\mathscr{E}}(t) = \underline{\hat{x}}\mathscr{E}_x(t) + \underline{\hat{y}}\mathscr{E}_y(t) + \underline{\hat{z}}\mathscr{E}_z(t) \quad \left(\text{assumed to be sinusoidal}\right)$$

$$= \underline{\hat{x}}A_x \cos(\omega t + \phi_x) + \underline{\hat{y}}A_y \cos(\omega t + \phi_y) + \underline{\hat{z}}A_z \cos(\omega t + \phi_z)$$

Convert to phasor domain:

$$\underline{\mathscr{E}}(t) = \mathrm{Re}\left(\underline{\hat{x}}A_x e^{j\phi_x} e^{j\omega t}\right) + \mathrm{Re}\left(\underline{\hat{y}}A_y e^{j\phi_y} e^{j\omega t}\right) + \mathrm{Re}\left(\underline{\hat{z}}A_z e^{j\phi_z} e^{j\omega t}\right)$$

$$= \mathrm{Re}\left\{\left(\underline{\hat{x}}A_x e^{j\phi_x} + \underline{\hat{y}}A_y e^{j\phi_y} + \underline{\hat{z}}A_z e^{j\phi_z}\right) e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{\left(\underline{\hat{x}}E_x + \underline{\hat{y}}E_y + \underline{\hat{z}}E_z\right) e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{\underline{E}\, e^{j\omega t}\right\}$$

Question:
Why does the frequency have to be the same for all components?

where

$$\underline{E} \equiv \underline{\hat{x}}E_x + \underline{\hat{y}}E_y + \underline{\hat{z}}E_z \qquad \text{Complex vector!}$$

Complex Vectors (cont.)

$$\underline{\mathscr{E}}(t) = \underline{\hat{x}}\mathscr{E}_x(t) + \underline{\hat{y}}\mathscr{E}_y(t) + \underline{\hat{z}}\mathscr{E}_z(t) \quad (\text{sinusoidal})$$

We have proven:

$$\underline{\mathscr{E}}(t) = \operatorname{Re}\left\{\underline{E}\, e^{j\omega t}\right\}$$

We thus work with phasor vectors the same way as we do with phasor scalars!

where

$$\underline{E} \equiv \underline{\hat{x}}E_x + \underline{\hat{y}}E_y + \underline{\hat{z}}E_z$$

Complex vector!

Notation:

$$\underline{\mathscr{E}}(t) \Leftrightarrow \underline{E}$$

$$\mathscr{E}_x(t) \Leftrightarrow E_x$$

$$\mathscr{E}_y(t) \Leftrightarrow E_y$$

$$\mathscr{E}_z(t) \Leftrightarrow E_z$$

Example 1.15 (Shen & Kong)

Assume $\underline{A} = \hat{x} + j\hat{y}$

Find the corresponding time-domain vector.

Note: $\left|\underline{\mathscr{A}}(t)\right| = \sqrt{\mathscr{A}_x^2(t) + \mathscr{A}_y^2(t)} = \sqrt{\cos^2 \omega t + \sin^2 \omega t} = 1$

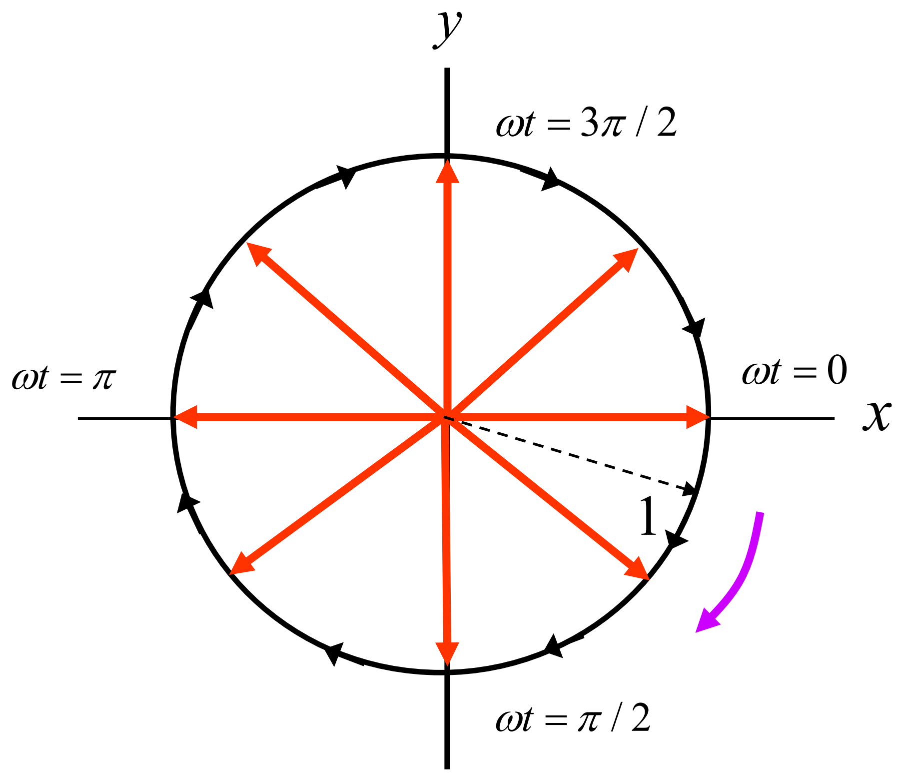
$$\underline{\mathscr{A}}(t) = \mathrm{Re}\left\{\underline{A}\, e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{(\underline{\hat{x}} + j\underline{\hat{y}})\, e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{(\underline{\hat{x}} + j\underline{\hat{y}})(\cos\omega t + j\sin\omega t)\right\}$$

$$= \underline{\hat{x}}\cos\omega t - \underline{\hat{y}}\sin\omega t$$

$$\underline{\mathscr{A}}(t) = \underline{\hat{x}}\cos\omega t - \underline{\hat{y}}\sin\omega t$$



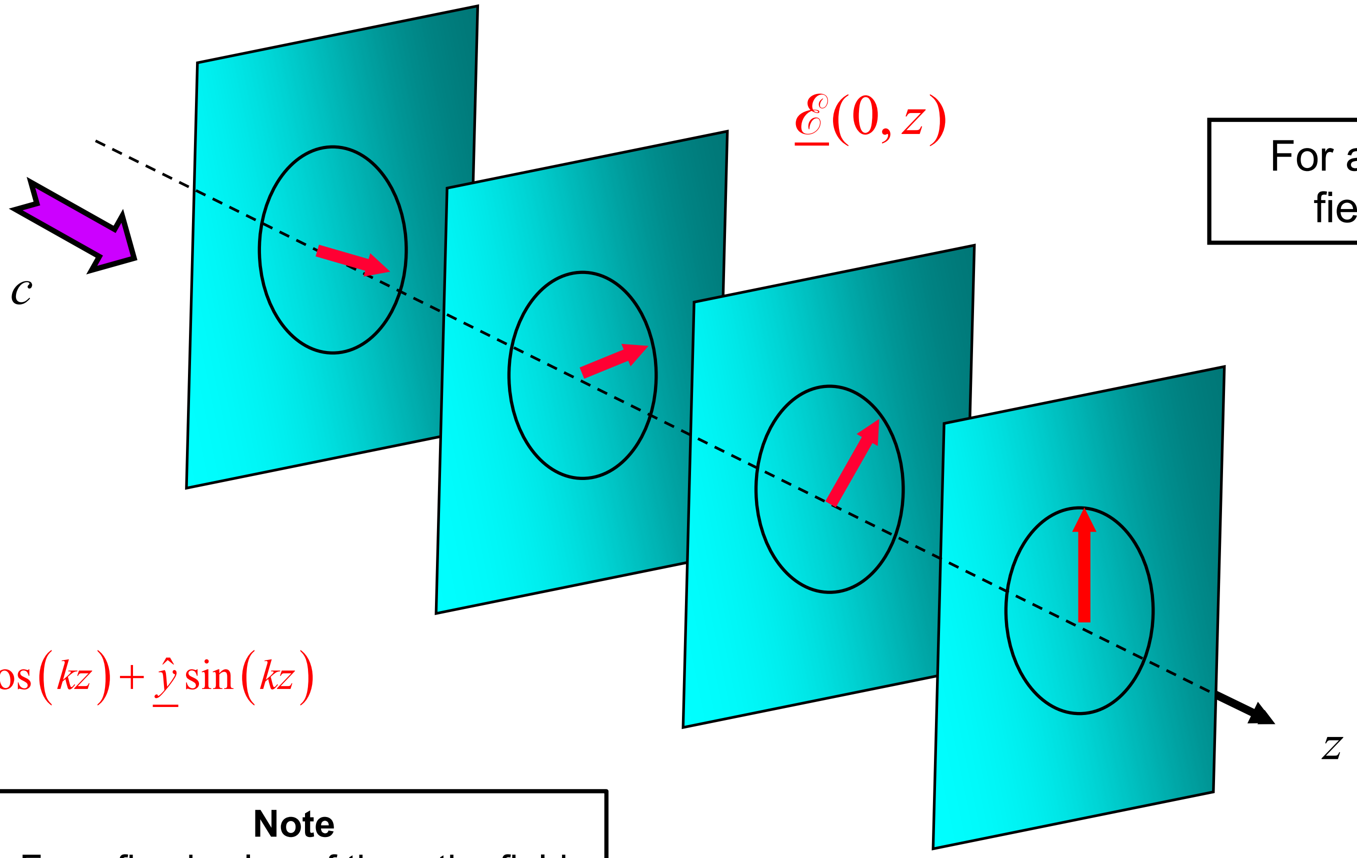
The vector rotates (clockwise) with time!

Example 1.15 (cont.)

Practical application:

A *circular-polarized plane wave* (discussed later)

$$\underline{\mathscr{E}}(t,z)=\underline{\hat{x}}\cos\left(\omega t-kz\right)-\underline{\hat{y}}\sin\left(\omega t-kz\right)$$



For a fixed value of <u>position</u> z, the electric field vector rotates clockwise in time.

$$\underline{\mathscr{E}}(t,0)=\underline{\hat{x}}\cos\left(\omega t\right)-\underline{\hat{y}}\sin\left(\omega t\right)$$

$$\underline{\mathscr{E}}(0,z)=\underline{\hat{x}}\cos\left(kz\right)+\underline{\hat{y}}\sin\left(kz\right)$$

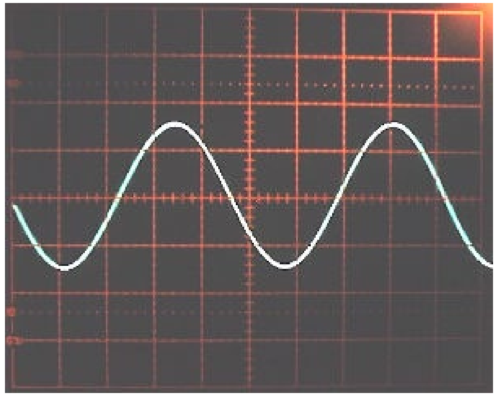
Note

For a fixed value of <u>time</u>, the field rotates counterclockwise in space.

This is a "snapshot" of the electric field at $t=0$.

Time Average of Time-Harmonic Quantities

$$v(t) = A\cos(\omega t + \phi)$$



$$\langle v(t) \rangle \equiv \frac{1}{T}\int_0^T A\cos(\omega t + \phi)\, dt = 0$$

$$\left(T = \frac{1}{f} = \frac{2\pi}{\omega} \right)$$

Note : $\cos^2 x = \left[1 + \cos\left(2x\right)\right]/2$

$$\langle v^2(t) \rangle = \frac{1}{T}\int_0^T A^2 \cos^2(\omega t + \phi)\, dt$$

$$\langle v^2(t) \rangle = \frac{A^2}{T}\int_0^T \left\{ \frac{1 + \cos[2(\omega t + \phi)]}{2} \right\} dt$$

Sinusoidal (time ave $= 0$)

$$\langle v^2(t) \rangle = \frac{A^2}{T}\left(\frac{T}{2}\right) = \frac{A^2}{2}$$

Hence

$$\langle v^2(t) \rangle = \frac{A^2}{2}$$

(Note: The average value of $\cos^2$ is $1/2$.)

Time Average of Time-Harmonic Quantities (cont.)

Next, consider the time average of a product of sinusoids:

$$\langle v(t)i(t)\rangle = \frac{1}{T}\int_0^T [A\cos(\omega t+\alpha)][B\cos(\omega t+\beta)]dt$$

Note : $\cos x \cos y = \left[\cos(x-y)+\cos(x+y)\right]/2$

$$\langle v(t)i(t)\rangle = \frac{1}{T}AB\int_0^T \cos(\omega t+\alpha)\cos(\omega t+\beta)dt$$

$$= \frac{AB}{T}\int_0^T\left[\frac{\cos(\alpha-\beta)+\cos(2\omega t+\alpha+\beta)}{2}\right]dt$$

$$= \frac{AB}{T}\left[T\,\frac{\cos(\alpha-\beta)}{2}\right]$$

Sinusoidal (time ave $= 0$)

$$= AB\left[\frac{\cos(\alpha-\beta)}{2}\right]$$

Time Average of Time-Harmonic Quantities (cont.)

Next, consider

$$VI^* = \left(A e^{j\alpha}\right)\left(B e^{j\beta}\right)^*$$
$$= \left(A e^{j\alpha}\right)\left(B e^{-j\beta}\right)$$
$$= AB\, e^{j(\alpha-\beta)}$$

Hence,

$$\mathrm{Re}\left(VI^*\right) = AB\cos\left(\alpha-\beta\right)$$

Recall that

$$\langle v(t)i(t)\rangle = AB\left[\frac{\cos\left(\alpha-\beta\right)}{2}\right]$$ (from previous slide)

Hence $$\langle v(t)i(t)\rangle = \frac{1}{2}\mathrm{Re}\left(VI^*\right)$$

Question:
Can we put the conjugate on the V instead of the I?

Time Average of Time-Harmonic Quantities (cont.)

The results directly extend to vectors that vary sinusoidally in time.

Consider: $$\underline{\mathscr{D}}(t)\cdot\underline{\mathscr{E}}(t) = \left(\mathscr{D}_x\mathscr{E}_x + \mathscr{D}_y\mathscr{E}_y + \mathscr{D}_z\mathscr{E}_z\right)$$

$$\mathscr{D}_{x,y,z}(t) = \mathrm{Re}\left[D_{x,y,z}e^{j\omega t}\right] \text{ etc.}$$

$$\begin{aligned}\left\langle \underline{\mathscr{D}}\cdot\underline{\mathscr{E}}\right\rangle &= \left\langle \mathscr{D}_x\mathscr{E}_x + \mathscr{D}_y\mathscr{E}_y + \mathscr{D}_z\mathscr{E}_z\right\rangle = \left\langle \mathscr{D}_x\mathscr{E}_x\right\rangle + \left\langle \mathscr{D}_y\mathscr{E}_y\right\rangle + \left\langle \mathscr{D}_z\mathscr{E}_z\right\rangle \\ &= \frac{1}{2}\mathrm{Re}\left(D_xE_x^*\right) + \frac{1}{2}\mathrm{Re}\left(D_yE_y^*\right) + \frac{1}{2}\mathrm{Re}\left(D_zE_z^*\right) \\ &= \frac{1}{2}\mathrm{Re}\left(D_xE_x^* + D_yE_y^* + D_zE_z^*\right)\end{aligned}$$

Hence $$\left\langle \underline{\mathscr{D}}(t)\cdot\underline{\mathscr{E}}(t)\right\rangle = \frac{1}{2}\mathrm{Re}\left(\underline{D}\cdot\underline{E}^*\right)$$

Time Average of Time-Harmonic Quantities (cont.)

The result holds for both dot product and cross products.

$$\left\langle \underline{\mathscr{D}}(t) \cdot \underline{\mathscr{E}}(t) \right\rangle = \frac{1}{2} \mathrm{Re}\left(\underline{D} \cdot \underline{E}^{*} \right)$$

$$\left\langle \underline{\mathscr{E}}(t) \times \underline{\mathscr{H}}(t) \right\rangle = \frac{1}{2} \mathrm{Re}\left(\underline{E} \times \underline{H}^{*} \right)$$

where

$$\underline{\mathscr{D}}(x, y, z, t) = \mathrm{Re}\left[\underline{D}(x, y, z) e^{j\omega t} \right]$$ etc.

Time Average of Time-Harmonic Quantities (cont.)

To illustrate, consider the time-average stored electric energy density [J/m^3] for a sinusoidal electric field.

$$U_E(t) = \frac{1}{2}\underline{\mathscr{D}}(t)\cdot\underline{\mathscr{E}}(t) \qquad \text{(from ECE 3318)}$$

$$\langle U_E(t)\rangle = \frac{1}{2}\left\langle \underline{\mathscr{D}}(t)\cdot\underline{\mathscr{E}}(t)\right\rangle$$

$$= \frac{1}{2}\left[\frac{1}{2}\mathrm{Re}\left(\underline{D}\cdot\underline{E}^*\right)\right]$$

$$\boxed{\langle U_E(t)\rangle = \frac{1}{4}\mathrm{Re}\left(\underline{D}\cdot\underline{E}^*\right)}$$

"Stored electric energy density"

Time Average of Time-Harmonic Quantities (cont.)

Similarly,

$$U_H(t) = \frac{1}{2}\underline{\mathscr{B}}(t) \cdot \underline{\mathscr{H}}(t)$$

"Stored magnetic energy density"

$$\boxed{\left\langle U_H(t) \right\rangle = \frac{1}{4}\mathrm{Re}\left(\underline{B} \cdot \underline{H}^*\right)}$$

$$\underline{\mathscr{S}}(t) \equiv \underline{\mathscr{E}}(t) \times \underline{\mathscr{H}}(t)$$

"Poynting vector"

$$\boxed{\left\langle \underline{\mathscr{S}}(t) \right\rangle = \frac{1}{2}\mathrm{Re}\left(\underline{E} \times \underline{H}^*\right)}$$